EC200D&ECx00G&EC600U&EG800G Series
QuecOpen(SDK) Ping API Reference Manual

LTE Standard Module Series

Version: 1.0

Date: 2023-09-08

Status: Released

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
http://www.quectel.com/support/sales.htm.

For technical support, or to report documentation errors, please visit:
http://www.quectel.com/support/technical.htm.
Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an "as available" basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2023. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2023-08-03	Chris PENG/ Herry GENG	Creation of the document
1.0	2023-09-08	Lambert ZHAO	First official release

Contents

Table Index

1 Introduction

Quectel ECx00G family, EC200D, EC600U and EG800G series modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see ***document [1]*** and ***document [2]***.

This document is applicable to QuecOpen® solution based on SDK build environment. It outlines Ping API, calling process and example on Quectel ECx00G family, EC200D, EC600U and EG800G series modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200D Series
-	EC600U Series
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
-	EG800G Series

NOTE

For the EC200D series module, the file paths mentioned in this document do not contain *components*.

2 Calling Process

Before calling the Ping interface, you need to register the network and perform data call to establish the data channel. These two steps can be skipped if the data channel has been established in other tasks. The calling process is shown as below.

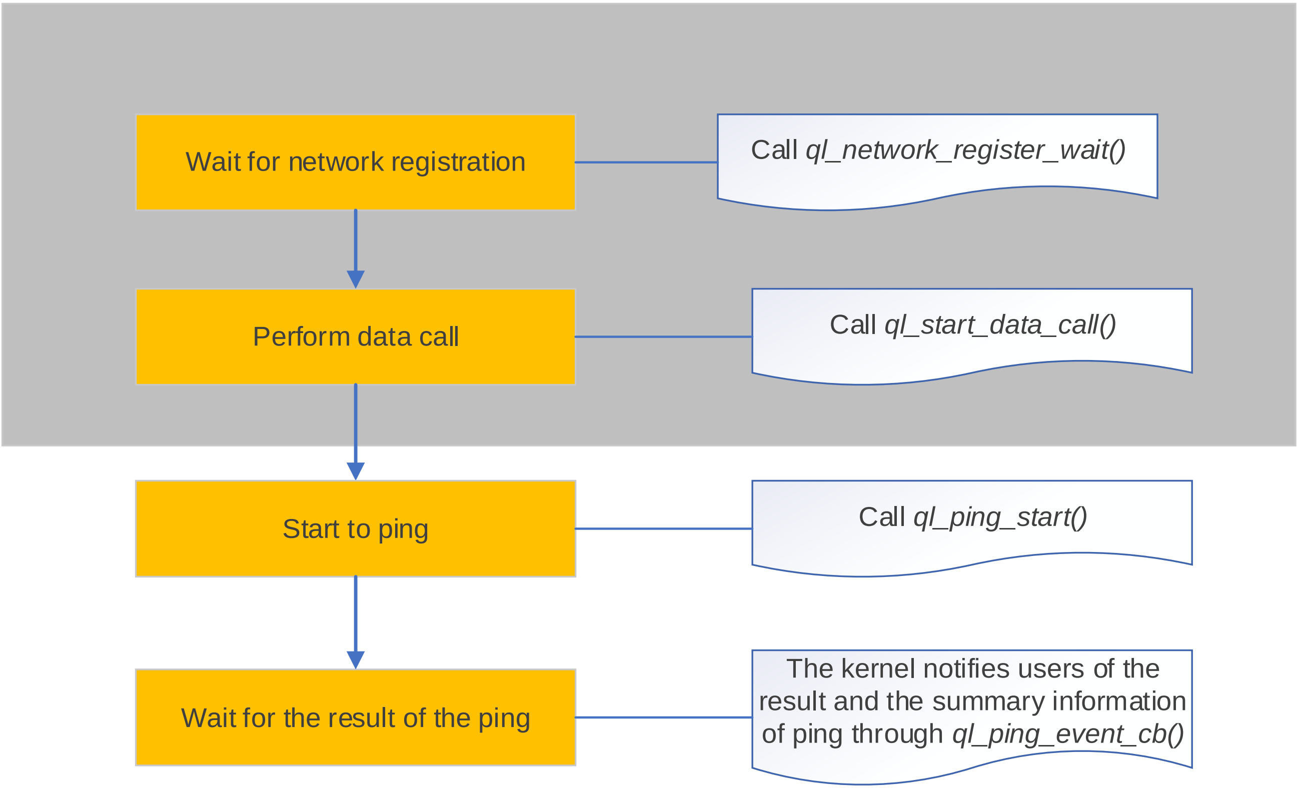

Figure 1: Ping Interface Calling Process

NOTE

See ***document [3]*** for details on *ql_network_register_wait()* and *ql_start_data_call()*.

3 Ping API

3.1. Header File

ql_ping_app.h, the header file of Ping API, is in the *components\ql-kernel\inc* directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

3.2. API Description

3.2.1. ql_ping_start

This function enables the ping functionality and sets the packets to be sent according to the configurations.

- **Prototype**

```
ping_session_id ql_ping_start(int cid, int sim_id, const char *host, ql_ping_config_type *ping_options, ql_ping_event_cb cb_fcn)
```

- **Parameter**

cid:
[In] Data channel number.

sim_id:
[In] (U)SIM card number. This parameter is generally set to 0 if the module does not support dual-card function.
0 (U)SIM 1
1 (U)SIM 2

host:
[In] Remote server address pinged.

ping_options:
[In] Ping configuration. See ***Chapter 3.2.1.1*** for details.

cb_fcn:
[In] Callback function of ping event. See ***Chapter 3.2.1.2*** for details.

- **Return Value**

Ping session ID. See ***Chapter 3.2.1.2.1*** for details. Integer type. Range: 1–10. 0 indicates failed creation of a session.

3.2.1.1. ql_ping_config_type

The structure of ping configuration:

```
typedef struct
{
    uint32_t   num_data_bytes;
    uint32_t   ping_response_time_out;
    uint32_t   num_pings;
    uint32_t   ttl;
} ql_ping_config_type;
```

- **Parameter**

Type	Parameter	Description
uint32_t	*num_data_bytes*	Size of ping data package. Unit: byte.
uint32_t	*ping_response_time_out*	Timeout of waiting for ping response. Unit: ms.
uint32_t	*num_pings*	Number of pings.
uint32_t	*ttl*	TTL for the ping package.

3.2.1.2. ql_ping_event_cb

This is the callback function of ping event.

- **Prototype**

```
typedef void(*ql_ping_event_cb)(ping_session_id ping_sess_id, uint16_t event_id, int evt_code, void *evt_param)
```

- **Parameter**

ping_sess_id:

[In] Ping session ID. See ***Chapter 3.2.1.2.1*** for details.

event_id:
[In] Ping type. See ***Chapter 3.2.1.2.2*** for details.

evt_code:
[In] Result codes of ping. See ***Chapter 3.2.1.2.3*** for details.

evt_param:
[In] Parameter of ping event.
When *event_id* is *QL_PING_STATS*, see ***Chapter 3.2.1.2.4*** for the definition of the type structure of this parameter.
When *event_id* is *QL_PING_SUMMARY*, see ***Chapter 3.2.1.2.5*** for the definition of the type structure of this parameter.

- **Return Value**

None

3.2.1.2.1. ping_session_id

The ping session ID is defined below.

```
typedef int ping_session_id
```

3.2.1.2.2. ql_ping_event_type

The enumeration of ping type:

```
typedef enum{
    QL_PING_STATS          = 1,
    QL_PING_SUMMARY        = 2,
}ql_ping_event_type;
```

- **Member**

Member	Description
QL_PING_STATS	Result for single ping.
QL_PING_SUMMARY	Summary information of pings.

3.2.1.2.3. ql_ping_result_code

The enumeration of the result codes of ping:

```
typedef enum{
    QL_PING_OK                         = 0,
    QL_PING_ERR_INVALID_PARAM          = (QL_COMPONENT_LWIP_SOCKET << 16) | 552
    QL_PING_ERR_SOCKET_NEW_FAILURE     = (QL_COMPONENT_LWIP_SOCKET << 16) | 554
    QL_PING_ERR_SOCKET_BIND_FAILURE    = (QL_COMPONENT_LWIP_SOCKET << 16) | 556
    QL_PING_ERR_DNS_FAILURE            = (QL_COMPONENT_LWIP_SOCKET << 16) | 565
    QL_PING_ERR_TIMEOUT                = (QL_COMPONENT_LWIP_SOCKET << 16) | 569
}ql_ping_result_code;
```

- **Member**

Member	Description
QL_PING_OK	Successful execution.
QL_PING_ERR_INVALID_PARAM	Invalid parameter.
QL_PING_ERR_SOCKET_NEW_FAILURE	Failed to create a socket.
QL_PING_ERR_SOCKET_BIND_FAILURE	Failed to bind the socket.
QL_PING_ERR_DNS_FAILURE	Failed to parse domain name.
QL_PING_ERR_TIMEOUT	Execution timeout.

3.2.1.2.4. ql_ping_stats_type

The structure of single ping result:

```
typedef struct{
    uint32_t     ping_rtt;
    uint32_t     ping_size;
    uint32_t     ping_ttl;
    char         resolved_ip_addr[256];
}ql_ping_stats_type;
```

- **Parameter**

Type	Parameter	Description
uint32_t	*ping_rtt*	RTT value.

uint32_t	*ping_size*	Size of ping data package.
uint32_t	*ping_ttl*	TTL value for ping data package.
char	*resolved_ip_addr*	Destination address pinged.

3.2.1.2.5. ql_ping_summary_type

The structure of the summary information of pings:

```
typedef struct{
    uint32_t   min_rtt;
    uint32_t   max_rtt;
    uint32_t   avg_rtt;
    uint32_t   num_pkts_sent;
    uint32_t   num_pkts_recvd;
    uint32_t   num_pkts_lost;
}ql_ping_summary_type;
```

- **Parameter**

Type	Parameter	Description
uint32_t	*min_rtt*	Minimum RTT. Unit: ms.
uint32_t	*max_rtt*	Maximum RTT. Unit: ms.
uint32_t	*avg_rtt*	Average RTT. Unit: ms.
uint32_t	*num_pkts_sent*	Number of data package sent.
uint32_t	*num_pkts_recvd*	Number of response package received.
uint32_t	*num_pkts_lost*	Number of the data package lost.

4 Example

For examples of Ping functionality, see *ping_demo.c*, the Ping client demo file on the application layer, which is in the *\components\ql-application\ping* directory.

5 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[2] Quectel_EC200D_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[3] Quectel_EC200D&ECx00G&EC600U&EG800G_Series_QuecOpen(SDK)_Data_Call_API_Reference_ Manual

Table 3: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
DNS	Domain Name Server
IoT	Internet of Things
IP	Internet Protocol
Ping	Packet Internet Groper
RTOS	Real-Time Operating System
RTT	Round-Trip Time
SDK	Software Development Kit
TTL	Time to Live